AEGIS: One-Query Adversarial Diagnostics over the GNN Vulnerability Spectrum

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***Abstract*—Graph neural networks in safety-critical settings (drug interaction, fraud detection, power-grid contingency screening) are vulnerable to adversarial graph perturbations, yet none derives, from a single analysis, the three diagnostics a practitioner needs: a globally optimal first-order perturbation direction, per-edge equilibrium-sensitivity rankings, and per-node sensitivity radii. AEGIS produces all three from one object, the *constrained sensitivity matrix* S_c, computed matrix-free via a Neumann-series resolvent and randomized SVD that scale to N=7,650 on a single GPU. For the spectral-norm-constrained IGNN subclass we prove a closed-form three-regime characterisation with critical budget $\varepsilon_{\text{crit}}$=$(1-\kappa)/\left\|W\right\|_2$ that trained models satisfy with 2–4× margin. S_c extends to continuous-edge-weight message passing (GCN, SAGE, GIN, APPNP, IGNN, edge-weighted GAT[†]): the continuous-to-discrete bridge is positive in 29/33 architecture–dataset cells (p<10^{-5}) and reaches $\tau = +0.996 \pm 0.002$ (10 seeds) against brute-force N-1 on full-graph Amazon Photo (N=7,650), while the one-query SVD direction matches 50-step PGD attackers. Top-v_{ij} masking cuts $\sigma_1(S_c)$-damage 42±8% (vs. 11% random) and survives adaptive recomputation; on IEEE power flow S_c recovers N-1 rankings on case14–118 (P@10=0.66–0.81), competitive with industry LODF.**

***Index Terms*—Graph neural networks, adversarial robustness, constrained sensitivity, implicit graph neural networks, randomized SVD, power-flow contingency screening**

I. INTRODUCTION

Graph neural networks are now deployed in domains where structural errors carry tangible consequences: fraudulent accounts evade detection via perturbed transaction edges [1], drug-interaction models misfire under perturbed molecular graphs [2], and power grids can miss critical contingencies when their graph representations are structurally fragile [3]. A practitioner deploying a GNN faces a question current tools leave unanswered: *which edges, if perturbed, would cause predictions to fail, and by how much?*

Structural attacks [1], [4], [5], [6] give per-edge surrogate gradients but no continuous direction or per-node budgets; smoothing [7], [8], [9], [10] and IBP-style certifiers [11], [12] give per-node certificates but no edge rankings or direction; robust-architecture defenses [13], [14], [15] harden models without surfacing per-edge vulnerability; sober-look critiques [16], [17] flag evaluation pitfalls. AEGIS (Adversarial Evaluation of Graph Integrity via Sensitivity) produces all three from one analytical object, S_c (section IV)—"one query" meaning a single matrix-free S_c construction (one randomized SVD over the equilibrium resolvent), not three separate analyses. The closed-form regime characterisation (theorem 1) is established for contractive implicit GNNs and S_c extends to any GNN with continuous edge-weight-modulated message passing (section V-F); the threat model targets edge-deletion and weight-perturbation (section II).

Contributions. (1) Constrained sensitivity operator. S_c specialises equilibrium IFT sensitivity [18], [19] to *structural* edge perturbations via the projection P_c; one matrix-free query reads the SVD-optimal direction, per-edge rankings, and per-node radii (dense-matching σ_1 to 0.03% at N=200; scaling to N=7,650)—no prior object yields all three at once. **(2) Phase-transition theory.** A worst-case three-regime characterisation with closed-form critical budget $\varepsilon_{\text{crit}}$=$(1-\kappa)/\left\|W\right\|_2$ for the spectral-norm-constrained IGNN subclass, which trained IGNNs satisfy with 2–4× margin across our $\kappa_{\max}$ sweep, giving practitioners a provable structural-perturbation safety threshold ($\sim$ 6% accuracy cost; theorem 1 and section V-D). **(3) Empirical evaluation.** 9 datasets, 7 architectures, 4 domains (330 runs): four-quadrant attack comparison, head-to-head vs. GR-BCD/PR-BCD [5], AGNNCert [11], and LODF (table IV), continuous-to-discrete transfer (fig. 7), and a power-grid N-1 case study (section VI).

II. PRELIMINARIES AND THREAT MODEL

Let $G = (V, E, A)$ be an undirected graph with $N = |V|$ nodes and adjacency $A \in \mathbb{R}^{N\times N}$, node features $X \in \mathbb{R}^{N\times d_{\text{in}}}$, and symmetric normalized adjacency $\hat{A} = D^{-1/2}(A+I)D^{-1/2}$ (self-loops included). Throughout, ϕ is the activation, $\sigma_i(\cdot)$ the i-th singular value, and $\text{vec}(\cdot)$ the column-major vectorization. The sensitivity objects introduced below are: the unconstrained $S \in \mathbb{R}^{Nd\times N^2}$, constrained $S_c \in \mathbb{R}^{Nd\times|E|}$, unrolled $S_K \in \mathbb{R}^{Nd\times N^2}$ (theorem 8), and per-node block S_v; the contractivity κ=$\left\|J_z\right\|_2$, spectral radius $\rho(J_z)$, pseudospectral index $\eta \geq 1$, critical budget $\varepsilon_{\text{crit}}$, and per-node radius r_v.

IGNN and IFT. A DEQ-GNN [20], [21] iterates a weight-tied F_θ to a fixed point; the IGNN instantiation [22] uses

$$F(Z, A) = \phi(\hat{A}ZW^\top + X_{\text{proj}}), \tag{1}$$

with $W \in \mathbb{R}^{d\times d}$ spectral-normalized [23] so $\kappa \leq \left\|\hat{A}\right\|_2 \left\|W\right\|_2 < 1$, ReLU ϕ, and a learned projection X_{proj}. Training uses implicit differentiation through the fixed point [19], [24] with optional Jacobian regularization [25], and a linear head $f(z^*) = W_{\text{cls}}z^* + b$ produces class predictions. At equilibrium, $G(z^*, A)$=z^*−$F(z^*, A)$=0 defines z^* as an implicit function of A, with IFT sensitivity

$$\frac{\partial z^*}{\partial p} = (I - J_z)^{-1}\frac{\partial F}{\partial p}\bigg|_{z^*}, \tag{2}$$

where $\left\|(I - J_z)^{-1}\right\|_2 \leq 1/(1-\kappa)$ via the convergent Neumann series, and the pseudospectral index η=$\left\|(I - J_z)^{-1}\right\|_2 (1 -$

$\rho) \geq 1$ [26] measures the gap to the spectral-radius bound (with η=1 for normal J_z). The IFT has been used for hyperparameter optimization [27] and implicit differentiation [19], but not for adversarial structural analysis.

Threat model. A white-box attacker with full access to the trained IGNN perturbs the normalized adjacency

$$\hat{A}' = \hat{A} + \delta\hat{A}, \quad \left\|\delta\hat{A}\right\|_F \leq \varepsilon, \tag{3}$$

with $\delta\hat{A}$ symmetric ($\delta\hat{A}$=$\delta\hat{A}^\top$, preserving the undirected graph), edge-only ($[\delta\hat{A}]_{ij}$=0 for $(i,j) \notin E$, so no new edges), and continuous in $\mathbb{R}$. AEGIS targets edge-deletion / weight-perturbation; insertion attacks (Nettack-class [1], [4]) require enlarging the basis to $\bar{E} \supseteq E$, a natural follow-up that preserves the S_c construction. Discrete deletions connect to the continuous model through theorem 7. Unlike input-feature perturbation [28], [29], structural perturbation changes the operator itself: z^* depends on A through both $J_z = \mathrm{diag}(\phi')(\hat{A} \otimes W)$ and $J_A = \partial F/\partial \mathrm{vec}(A)$, motivating S_c in section III.

III. Adversarial Equilibrium Theory

Building on the threat model of section II, we derive the analytical machinery behind the three diagnostics promised in section I: per-edge rankings, the SVD-optimal direction, and per-node radii. The main result (theorem 1) characterizes how vulnerability evolves with perturbation magnitude, and the constrained sensitivity matrix S_c provides the unified object from which the three diagnostics follow.

Theorem 1 (Vulnerability Characterization for Contractive Implicit GNNs)**.** *Let $F_\theta(Z, \hat{A}) = \phi(\hat{A}ZW^\top + X_{\mathrm{proj}})$ be an IGNN-class operator satisfying:*

(A1) *ϕ is ReLU or any 1-Lipschitz activation; nonsmoothness at zero is handled via the conservative IFT [30] (proof below).*
(A2) *$\|W\|_2 \leq c$ via spectral normalization.*
(A3) *$\|J_z\|_2 \leq \kappa < 1$ verified post-training (κ=0.14–0.59 across all benchmarks in our suite; section V-E). (A2) does not imply (A3) automatically; we report κ alongside $\varepsilon_{\mathrm{crit}}$ so practitioners can audit, and* AEGIS *continues to return S_c rankings and SVD-optimal attacks if (A3) is not satisfied, with the formal guarantees below then scoped to the contractive regime.*

Define the critical budget

$$\varepsilon_{\mathrm{crit}} = \frac{1-\kappa}{\|W\|_2}. \tag{4}$$

Then structural perturbations $\delta\hat{A}$ with $\left\|\delta\hat{A}\right\|_F = \varepsilon$ exhibit three regimes:

(a) Subcritical *($\varepsilon < \varepsilon_{\mathrm{crit}}$): The perturbed operator remains contractive with a unique fixed point satisfying*

$$\|\Delta z^*\|_F \leq \sigma_1(S) \cdot \varepsilon + O(\varepsilon^2), \tag{5}$$

where $S = (I - J_z)^{-1}J_A$ is the structural sensitivity matrix and $\sigma_1(S) \leq \|J_A\|_{\mathrm{op}}/(1-\kappa)$.

(b) Critical *($\varepsilon \to \varepsilon_{\mathrm{crit}}$): The resolvent obeys the unconditional eigenvalue bound $\left\|(I - J_z')^{-1}\right\|_2 \geq 1/\min_i |1-\lambda_i(J_z')|$, so it blows up precisely when an eigenvalue of J_z' approaches 1—a vanishing spectral gap to* 1, not *$\|J_z'\|_2 \to 1$ on its own. The rate is $\Omega(1/(\varepsilon_{\mathrm{crit}}-\varepsilon))$ when J_z' is normal with a dominant real-positive eigenvalue (the Perron mode of symmetric $\hat{A}, W$); in general $\varepsilon_{\mathrm{crit}}$ is the norm-based contraction radius that* lower-bounds *the eigenvalue divergence threshold, the slack reflecting the nonnormality of J_z' (bounded by η, theorem 2).*

(c) Supercritical *($\varepsilon \geq \varepsilon_{\mathrm{crit}}$): The sufficient contraction certificate no longer applies ($\|J_z'\|_2$ may exceed 1) and the part-(a) first-order guarantees do not apply; the certificate is reported as a sufficient safety boundary, validated empirically by the $\kappa_{\max}$ sweep of section V-D (trained IGNNs hold with 2–4× margin).*

Proof. **(a).** Under (A1), F_θ is piecewise-affine on each ReLU linear region; the activation pattern at z^* is generically stable for sufficiently small $\left\|\delta\hat{A}\right\|$ (the boundary set $\{\hat{A} : z^*(\hat{A})$ lies on a region face$\}$ has Lebesgue measure zero), so the conservative-gradient calculus of Bolte–Pauwels [30] yields a conservative IFT on each region. With $J_z = \mathrm{diag}(\phi')(\hat{A} \otimes W)$ and $\kappa \leq \left\|\hat{A}\right\|_2 \|W\|_2$, applying IFT to $G(z,\hat{A})$=z−$F(z,\hat{A})$=0 at $(z^*, \hat{A})$ gives

$$\Delta z^* = (I - J_z)^{-1}J_A \cdot \mathrm{vec}(\delta\hat{A}) + O(\left\|\delta\hat{A}\right\|^2), \tag{6}$$

with $J_A = \partial F/\partial \mathrm{vec}(A)$ and Neumann $\left\|(I - J_z)^{-1}\right\|_2 \leq 1/(1-\kappa)$ yielding (5). For contractivity preservation, $\|J_z'\|_2 \leq (\left\|\hat{A}\right\|_2 + \varepsilon)\|W\|_2$, so $\varepsilon < \varepsilon_{\mathrm{crit}}$ guarantees $\|J_z'\|_2 < 1$.

(b)–(c). For any M, $(I-M)^{-1}$ has eigenvalues $(1-\lambda_i(M))^{-1}$, hence $\left\|(I-M)^{-1}\right\|_2 \geq \rho((I-M)^{-1}) = 1/\min_i |1-\lambda_i(M)|$ *unconditionally*. Along the worst-case direction $\|J_z'\|_2 \leq (\left\|\hat{A}\right\|_2 + \varepsilon)\|W\|_2$, so $1 - \|J_z'\|_2 \geq \|W\|_2(\varepsilon_{\mathrm{crit}}-\varepsilon)$ and, since $\rho \leq \|\cdot\|_2$, $\varepsilon < \varepsilon_{\mathrm{crit}}$ certifies $\|J_z'\|_2 < 1$. The certificate is one-sided: $\|J_z'\|_2 \to 1$ need not drive $\min_i |1-\lambda_i| \to 0$ (e.g. M=$\mathrm{diag}(-s, 0)$ has $\|M\|_2$=s yet $\left\|(I-M)^{-1}\right\|_2$=1). When J_z' is normal with dominant real-positive eigenvalue $\lambda^\star$=$\|J_z'\|_2$ (the Perron mode for symmetric $\hat{A}, W$), $\min_i |1-\lambda_i|$=$1-\lambda^\star$=$\|W\|_2(\varepsilon_{\mathrm{crit}}-\varepsilon)$ yields the $\Omega(1/(\varepsilon_{\mathrm{crit}}-\varepsilon))$ rate; a negative or complex dominant eigenvalue keeps $\min_i |1-\lambda_i|$ away from 0 (the same $\mathrm{diag}(-s, 0)$), so $\varepsilon_{\mathrm{crit}}$ only lower-bounds the threshold, slack controlled by η (theorem 2). For $\varepsilon \geq \varepsilon_{\mathrm{crit}}$ Banach contraction fails and the iteration may converge elsewhere, oscillate, or diverge. □

$\varepsilon_{\mathrm{crit}}$ is a closed-form sufficient bound that S_c tightens in practice, 7–14× over the unconstrained $\sqrt{r}$ bound on 50-node subgraphs (section V-A). The Neumann series requires the operator-norm κ rather than $\rho(J_z)$ [31]; the gap is the pseudospectral index η, bounded by the weight matrix alone:

Observation 2 (Graph-Independent Nonnormality Bound, all-active case)**.** *For $J_z = \mathrm{diag}(\phi')(\hat{A} \otimes W)$ with symmetric $\hat{A}$ and all activations positive ($\phi' \equiv 1$): $\eta \leq \kappa(V_W)$ where*

$W{=}V_W D_W V_W^{-1}$ *and* $\kappa(V_W){=}\|V_W\|_2 \left\|V_W^{-1}\right\|_2$*, independent of* $\hat{A}$.

Proof. $J_z{=}\hat{A} \otimes W$ has Kronecker eigenvalues $\lambda_i(\hat{A})\lambda_j(W)$ [31]; symmetric $\hat{A}$ gives $\kappa(U_{\hat{A}}){=}1$, so the joint eigenvector matrix has condition number $\kappa(V_W)$, and the diagonalisable bound $\left\|(I{-}M)^{-1}\right\|_2 \le \kappa(V)/(1{-}\rho(M))$ yields $\eta \le \kappa(V_W)$. □

Remark 3 (Empirical extension to general ReLU patterns)**.** *The proof above requires* $\phi' \equiv 1$*; for general ReLU patterns,* $\eta \in [1.19, 2.47]$ *on our suite (section V-E), tracking* $\kappa(V_W)$ *closely. Graph structure does not amplify nonnormality; the moderate* η *traces to spectral-norm regularisation of* W.

Proposition 4 (Maximally Sensitive First-Order Structural Perturbation)**.** *The perturbation* δA^* *maximizing* $\|\Delta z^*\|$ *to first order subject to* $\|\delta A\|_F \le \varepsilon$ *is* $\delta A^* = \varepsilon \cdot \mathrm{reshape}(v_1, N \times N)$*, where* v_1 *is the leading right singular vector of* S*. The maximum first-order equilibrium shift is* $\varepsilon \cdot \sigma_1(S)$.

By the variational characterisation of σ_1, the direction (theorem 4) is optimal for hidden-state damage within the pseudospectral envelope of theorems 2 and 3, and empirically aligns with classification damage (section V-B). The *constrained sensitivity matrix* $S_c \in \mathbb{R}^{Nd\times|E|}$ has columns $[S_c]_{:,k} = S_{:,iN+j}{+}S_{:,jN+i}$ paired with edge basis $b_k{=}(e_ie_j^\top{+}e_je_i^\top)/\sqrt{2}$ ($\|b_k\|_F{=}1$, so $\sigma_1(S_c)$ is consistent with $\left\|\delta\hat{A}\right\|_F{=}\varepsilon$). The $N^2 \to |E|$ projection P_c is the standard duplication-matrix reduction [32] on the edge-supported symmetric subspace; the operational contribution is the matrix-free $S_c v$ in algorithm 1. The first-order envelope identifies the dangerous direction rather than a global safety margin (table II).

Proposition 5 (Per-Node First-Order Sensitivity Radius)**.** *For a linear head* $f(z){=}Wz{+}b$ *and a node* v *classified as* y_v *with positive margins* $m_v^{(c)}{=}f_{y_v}(z_v^*){-}f_c(z_v^*){>}0$ *to every competitor* $c{\neq}y_v$*, the first-order sensitivity radius*

$$r_v = \min_{c\neq y_v} \frac{m_v^{(c)}}{\|(W_{y_v}{-}W_c)\,S_v\|_2} \tag{7}$$

preserves the classification of v *for any* $\left\|\delta\hat{A}\right\|_F < r_v$*, where* S_v *are the block-rows of* S *at node* v*; substituting* $S_{c,v}$ *gives the tighter constrained-threat radius* $r_v^{(c)}{\ge}r_v$*. Minimizing over* all *competitors is the distance to the nearest first-order boundary, so the bound holds even when the runner-up changes under perturbation; the cheaper runner-up surrogate* $\hat{r}_v{=}m_v/(\|W_{y_v}{-}W_{c^*}\|_2\,\|S_v\|_2)$ *(*$c^*{=}\arg\max_{c\neq y_v} f_c$*) can be optimistic when a non-runner-up class is more aligned.*

Proof sketch. By Thm. 1(a), $\Delta z_v^* \approx S_v\,\mathrm{vec}(\delta A)$, so for each competitor $|\Delta(f_{y_v}{-}f_c)| \le \|(W_{y_v}{-}W_c)S_v\|_2\,\|\delta A\|_F$ by Cauchy–Schwarz; the prediction survives while every margin stays positive, i.e. $\|\delta A\|_F < m_v^{(c)}/\|(W_{y_v}{-}W_c)S_v\|_2$ for all c, whose minimum is r_v. Intuitively, r_v is the smallest margin divided by two amplification factors (decision-boundary sensitivity to hidden-state change, and equilibrium sensitivity to structure); high-margin, low-sensitivity nodes get large radii.

Remark 6 (Sensitivity threshold semantics)**.** r_v *is a first-order sensitivity threshold, not a probabilistic certificate: it indicates the perturbation scale at which the linearization predicts a classification change for node* v*, and is locally tight for small* ε *but can be violated at larger magnitudes where higher-order terms dominate. The two approaches are complementary: randomized smoothing [7] certifies which nodes are guaranteed safe, while* AEGIS *differentiates which edges and nodes are most sensitive and in which direction (section V-B). Empirically, breach rate is* 0% *at* $\varepsilon{=}0.01$ *and rises with* ε *as expected for first-order approximations (section V-B).*

A. Continuous-to-Discrete Transfer

Proposition 7 (Continuous-to-Discrete Ranking Transfer)**.** *Under (A1)–(A3), let* $w_k = [\hat{A}]_{ij}$ *denote the normalized weight of edge* $k = (i,j) \in E$*,* $v_k = \|[S_c]_{:,k}\|_2$ *the continuous vulnerability score, and* $d_k = \|z^*(A) - z^*(A \setminus k)\|_2$ *the discrete damage from fixed-normalization removal of edge* k*. Assume all single-edge removals are subcritical:* $\sqrt{2}\max_k w_k < \varepsilon_{\mathrm{crit}}$.
(a) First-order bridge. *The discrete damage satisfies*

$$d_k = w_k \cdot v_k + R_k, \quad |R_k| \le \frac{L_J}{2(1-\kappa)^2} \cdot w_k^2, \tag{8}$$

where $L_J \le \|W\|_2^2\,\|z^*\|$ *is the second-order curvature constant of* $z^*(A)$*, finite at the fixed point.*
(b) Ranking preservation. *For two edges* k_1, k_2 *with* $v_{k_1} > v_{k_2}$ *and* $w_{k_1} \ge w_{k_2}$*, we have* $d_{k_1} > d_{k_2}$ *whenever*

$$w_{k_1} < \frac{v_{k_1} - v_{k_2}}{L_J/(1-\kappa)^2}. \tag{9}$$

Under uniform weights ($w_k = w$ *for all* k*),* $v_{k_1} > v_{k_2}$ *implies* $d_{k_1} > d_{k_2}$ *whenever* $w < \min_{k_1\neq k_2}(v_{k_1} - v_{k_2}) \cdot (1-\kappa)^2/L_J$.

Proof. **(a).** We model deletion as *fixed-normalization* masking, $[\delta\hat{A}]_{ij}{=}[\delta\hat{A}]_{ji}{=}{-}w_k$ with the degree matrix D held fixed, matching the continuous edge-weight perturbation, so by the S_c construction $\|\Delta z^*\| \approx w_k v_k$. (recompute-normalization adds an $O(d_i^{-1})$ incident-edge rescaling, negligible off low-degree nodes; agreement in section V-A.) For the remainder, the IFT gives $\partial z^*/\partial\mathrm{vec}(A){=}(I{-}J_z)^{-1}J_A$, and the dominant curvature term $(I{-}J_z)^{-1}(\partial J_z/\partial\mathrm{vec}(A))(I{-}J_z)^{-1}J_A$ carries two resolvents $(1{-}\kappa)^{-1}$, two factors $\|W\|_2$, and the equilibrium norm $\|z^*\|$ (since $\partial J_z/\partial\mathrm{vec}(A) \propto W$ and $J_A \propto z^*W^\top$), so $L_J \le \|W\|_2^2\,\|z^*\|$, finite since $\|z^*\| \le \|X_{\mathrm{proj}}\|/(1{-}\left\|\hat{A}\right\|_2\|W\|_2)$ (from $\|\phi(u)\|{\le}\|u\|$ at the fixed point and the spectral normalization $\left\|\hat{A}\right\|_2\|W\|_2{<}1$ of section II), giving $\left\|\partial^2 z^*/\partial\mathrm{vec}(A)^2\right\|_2 \le (1{-}\kappa)^{-2}L_J$. The path meets finitely many ReLU regions (simultaneous crossings nongeneric, measure zero); summing the second-order remainder over these regions, with the conservative IFT [30] on each, gives $|R_k| \le L_J w_k^2/(2(1{-}\kappa)^2)$. **(b).** From (a), $d_{k_1}{-}d_{k_2} \ge w_{k_1}v_{k_1}{-}w_{k_2}v_{k_2}{-}C(w_{k_1}^2{+}w_{k_2}^2)$ with $C{=}L_J/(2(1-\kappa)^2)$; under $v_{k_1}{>}v_{k_2}$, $w_{k_1}{\ge}w_{k_2}$, the first term is positive and (9) ensures the first-order gap dominates. □

Empirically $|R_k|$ is 2–10× smaller than $L_J w_k^2$; the constrained projection S_c turns the 0.31 vacuous bound into a tight one (1.00; section V-A) [31], [26]. The edge-weighted score $w_k v_k$ predicted by (a) reaches $\tau = +\mathbf{0.996}$ against brute-force N-1 on full-graph Amazon Photo (10 seeds, section V-F), corroborating the first-order bridge at scale.

B. Generalization to Explicit GNNs

Observation 8 (Structural Sensitivity for K-Layer GNNs)**.** *Let $Z_K = g_K \circ \cdots \circ g_1(X, A)$ be a K-layer GNN where each layer g_l is differentiable with respect to both its input and A. Define the* unrolled sensitivity matrix

$$S_K = \frac{\partial \mathrm{vec}(Z_K)}{\partial \mathrm{vec}(A)} = \sum_{l=1}^{K} \left(\prod_{k=l+1}^{K} J_z^{(k)} \right) J_A^{(l)}, \tag{10}$$

where $J_z^{(k)} = \partial g_k / \partial Z_{k-1}$ and $J_A^{(l)} = \partial g_l / \partial \mathrm{vec}(A)$. Then:
***(a)** The first-order shift bound $\|\Delta Z_K\|_F \le \sigma_1(S_K) \cdot \|\delta A\|_F + O(\|\delta A\|^2)$ holds, with*

$$\sigma_1(S_K) \le \sum_{l=1}^{K} \left(\prod_{k=l+1}^{K} \left\| J_z^{(k)} \right\|_2 \right) \left\| J_A^{(l)} \right\|_2 . \tag{11}$$

***(b)** The constrained construction S_c and per-node radius r_v (theorem 5) apply identically with S_K in place of S.*

For weight-tied models, $\sigma_1(S_K) \le \|J_A\|_2 (1-\kappa^K)/(1-\kappa)$ converges to the IGNN bound as $K \to \infty$. Explicit GNNs inherit the full computational pipeline with empirical tightness 0.99–1.02 across the architecture suite (section V-F); the closed-form $\varepsilon_{\mathrm{crit}}$ is available only for the contractive subclass.

IV. AEGIS CONSTRUCTION

The pipeline turns section III into a practical tool through matrix-free computation, sidestepping both the materialisation of $S \in \mathbb{R}^{Nd \times N^2}$ and the formation of the resolvent. One S_c computation returns the per-edge spectrum $\{v_{ij}\}$, the SVD-optimal direction $\delta\hat{A}^\star$, and per-node radii $\{r_v\}$, plus $\varepsilon_{\mathrm{crit}}$ and the regime characterisation of theorem 1 for IGNN-class operators. A dense path ($N \le 200$) computes an exact SVD; a matrix-free path ($N > 200$, validated to N=7,650) uses JVP/VJP queries and randomized SVD. Figure 1 illustrates the four stages; algorithm 1 summarises the matrix-free routing.

Matrix-free operator. $S_c v = (I - J_z)^{-1} J_A P_c v$ is evaluated via the truncated Neumann series $\sum_{k=0}^{K} J_z^k b$ with depth K adaptive in κ (early stop at $\left\| J_z^k b \right\| < 10^{-6} \|b\|$); each Neumann term and $J_A \,\mathrm{vec}(\delta A)$ are forward-mode JVPs in $O(Nd)$ time. P_c is applied implicitly, so neither S_c nor S is formed; the dense path ($N \le 200$) materialises J_z, J_A and applies a direct linear solve. The rSVD [33] (k=p=10, n_{iter}=2) returns (u_1, σ_1, v_1) for the direction $\delta\hat{A}^\star$=ε·$\mathrm{sym}(P_c v_1)$ (theorem 4); column norms give $\{v_{ij}\}$, and per-node radii follow theorem 5 via the node-restricted RMATVEC. The well-separated singular spectrum (gap $(\sigma_1 - \sigma_2)/\sigma_1$=0.39–0.50; fig. 2) is why one rSVD query matches iterative attackers (section V-B).

Cost. Each $S_c v$ is $O(K \cdot Nd)$ time and $O(Nd)$ memory ($K \in [20, 50]$ for $\kappa < 0.8$); removing the $O((Nd)^3)$ solve lifts the previous $N \approx 300$ ceiling (section V-D).

Algorithm 1 AEGIS Vulnerability Analysis (matrix-free). Assumes the equilibrium z^* is pre-computed on the analysis (sub)graph. Dense fallback ($N \le 200$): form $S = (I - J_z)^{-1} J_A$ and run a deterministic SVD in place of lines 2–11.

Require: F, z^*, $\hat{A}$, budget ε; SVD rank k, oversampling p, power iters n_{iter}; Neumann tol. τ; edge set E; (opt.) head h, logits Y, labels y.
Ensure: $\sigma_1(S_c)$, $\delta\hat{A}^\star$, $\{v_{ij}\}$, $\{r_v\}$, $\varepsilon_{\mathrm{crit}}$.

1: $\kappa \leftarrow \|J_z\|_2$ via power iteration; $K \leftarrow \lceil \log(1/\tau) / \log(1/\kappa) \rceil$.
2: MATVEC(v) := $\sum_{j=0}^{K} J_z^j \mathrm{JVP}_F^{\hat{A}}(P_c v)|_{z^*}$; RMATVEC($u$) := $P_c^\top \mathrm{VJP}_F^{\hat{A}}(\sum_{j=0}^{K} (J_z^\top)^j u)|_{z^*}$.
3: Draw $\Omega \sim \mathcal{N}(0, I) \in \mathbb{R}^{|E| \times (k+p)}$; $Y \leftarrow$ MATVEC(Ω) column-wise.
4: **for** $j = 1, \ldots, n_{\mathrm{iter}}$ **do**
5: $\quad Y \leftarrow \mathrm{QR}$(MATVEC(RMATVEC($Y$)))
6: **end for**
7: $Q \leftarrow \mathrm{QR}(Y)$; $B^\top \leftarrow$ RMATVEC(Q); $(\widetilde{U}, \Sigma, V^\top) \leftarrow \mathrm{SVD}(B)$; $U \leftarrow Q\widetilde{U}$ {rSVD [33]}
8: $\sigma_1 \leftarrow \Sigma_{1,1}$; $\delta\hat{A}^\star \leftarrow \varepsilon \cdot \mathrm{sym}(P_c V_{:,1}) / \|\mathrm{sym}(P_c V_{:,1})\|_2$.
9: $v_{ij} \leftarrow \|$MATVEC$(e_{ij})\|_2$ for each $(i, j) \in E$ {per-edge spectrum}
10: **if** h, Y, y supplied **then**
11: $\quad r_v \leftarrow \min_{c \ne y_v} m_v^{(c)} / \|(W_{y_v} - W_c)\, S_{c,v}\|_2$ for each v {Prop. 5}
12: **end if**
13: **if** $\|W\|_2$ extractable **then** $\varepsilon_{\mathrm{crit}} \leftarrow (1-\kappa)/\|W\|_2$ {Thm. 1}
14: **return** $(\sigma_1, \delta\hat{A}^\star, \{v_{ij}\}, \{r_v\}, \varepsilon_{\mathrm{crit}})$.

V. EXPERIMENTS

Setup. We evaluate AEGIS on 9 datasets across 4 domains with 10 random seeds each (PyTorch [34]). IGNN [22] uses $F(Z)$=ReLU$(\hat{A}ZW^\top + U(X))$ with $W \in \mathbb{R}^{64 \times 64}$ spectral-normalised [23] and Adam (lr 0.01); ContractiveGCN-PF mirrors this with 5 bus features and 2-head readout. The spectral-norm constraint costs ∼6% Cora accuracy (77.5% vs. ∼83%). Datasets: Cora, Citeseer, Pubmed [35], Amazon Photo [36], WikiCS [37], IEEE 14/30/57/118 [38], [39]. Baselines: smoothing (Gaussian σ=0.01, 200 MC, Clopper–Pearson α=0.001), Mettack with Meta-Self GCN surrogate, LODF as $\mathrm{PTDF}_{ij}/(1 - \mathrm{PTDF}_{jj})$ from the DC B-matrix [40, Ch. 11]. The 9 datasets and 7 architectures are cumulative: all seven architectures appear only in table V and fig. 7 (33 cells); other studies use IGNN, and power flow the four IEEE cases.

A. Cross-Domain Adversarial Analysis

Table I verifies the contractivity assumption (A3) on every benchmark (κ<1) and reports the per-dataset closed-form $\varepsilon_{\mathrm{crit}}$ alongside an attack advantage of 3.2–4.1× over random perturbations. Unless otherwise stated, IGNN experiments in section V-A–section V-E use 50-node BFS ego-subgraphs centred on the highest-degree node and report *local vulnera-*

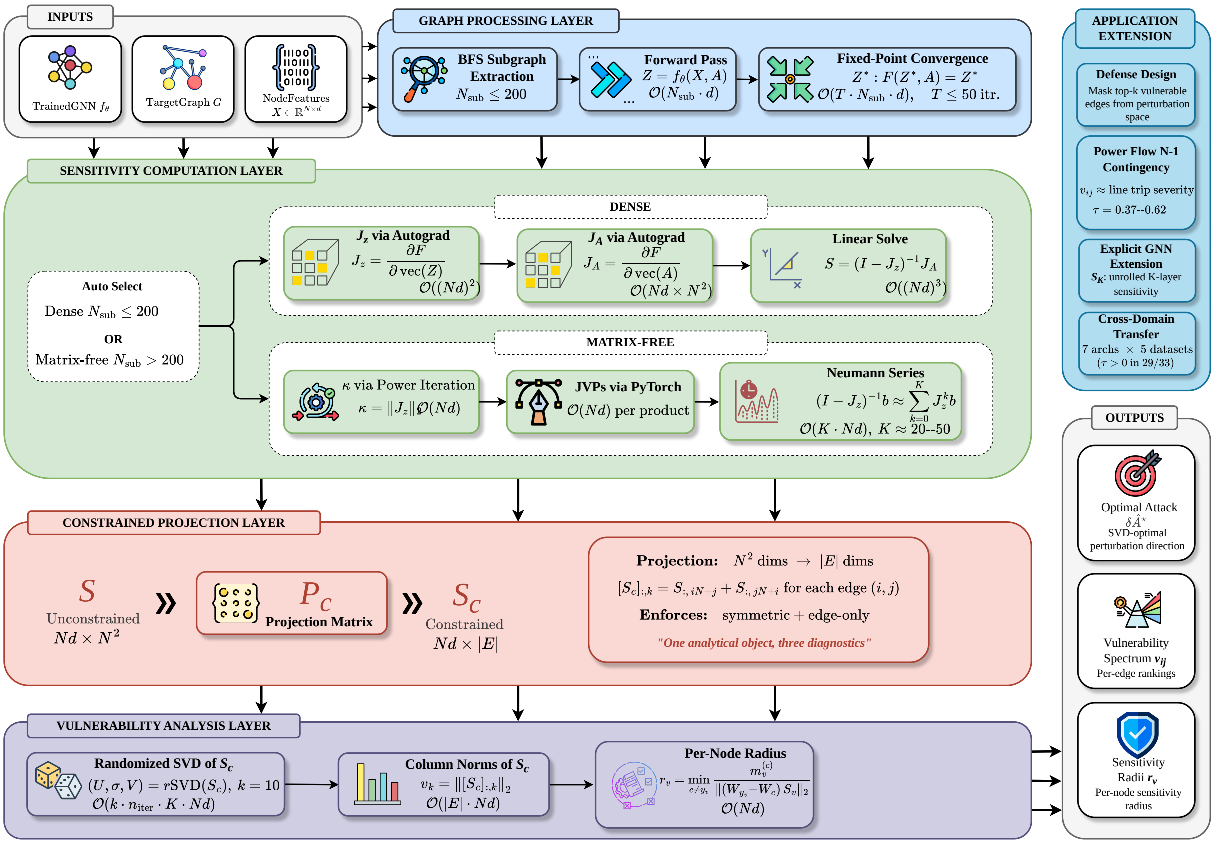


Fig. 1: The four-stage AEGIS pipeline: (1) graph processing (BFS subgraph or full-graph forward / fixed-point pass); (2) sensitivity computation (dense Jacobian for $N \leq 200$, matrix-free Neumann/JVP otherwise); (3) constrained projection $N^2 \to |E|$ via P_c; (4) vulnerability analysis (randomised SVD, S_c column norms, per-node radii). The matrix-free path queries S_c via JVPs/VJPs without materialising it, scaling to N=7,650.

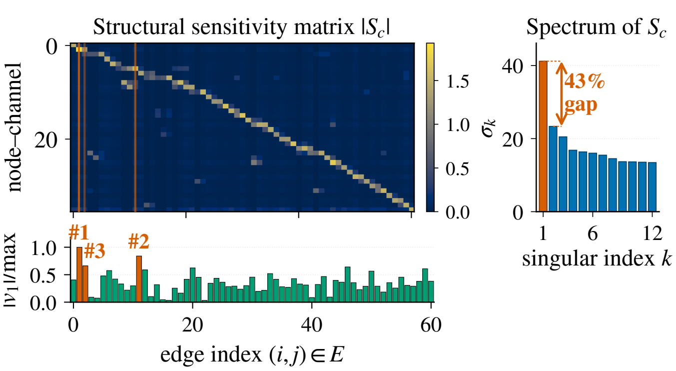


Fig. 2: $|S_c|$ on a 50-node Cora ego-graph (IGNN, N=50, d=16, $|E|$=61, seed 0). Vermilion lines: top-3 edges by $|v_1|$ (SVD-optimal direction, bar strip). Sidebar: singular spectrum, σ_1=41.2, 43% gap to σ_2 (cross-benchmark $(\sigma_1-\sigma_2)/\sigma_1 \in [0.39, 0.50]$; section V-B).

bility; graph-level assessment uses the matrix-free pipeline of section V-D. All bounds use the operator-norm $\kappa = \|J_z\|_2$.

The envelope ratio is measured on equilibrium shift $\|\Delta z^*\|_F$ (table II), distinct from the prediction-flip (breach) rate of fig. 4 (flips stay $< 2\%$ at $\varepsilon \leq 0.10$, $< 6\%$ at ε=0.20).

Heuristic and surrogate baselines. AEGIS beats degree-proportional, edge-betweenness, and spectral rankings at ε=0.01 (+6–148% AtkAdv, Wilcoxon p<0.001) and inflicts 3–10× more equilibrium damage than Mettack [4] at the early-warning regime $k \in \{1, \ldots, 5\}$ (**149/150** paired wins, p<10^{-43}); GR-BCD/PR-BCD [5] are in table IV. The distinctive value appears under discrete removal (fig. 3).

TABLE I: Per-dataset IGNN characterisation (10 seeds). $\kappa = \|J_z\|_2$ verifies (A3); $\varepsilon_{\text{crit}} = (1-\kappa)/\|W\|_2$ is the theoretical critical budget. AtkAdv: AEGIS damage / random damage. Cov%: fraction of subgraph nodes with positive first-order radius r_v.

Dataset	Test Acc	κ	$\varepsilon_{\text{crit}}$	AtkAdv	Cov%
Cora	$77.5_{\pm 1.7}$	$.33_{\pm .14}$	$.66_{\pm .15}$	$3.6_{\pm .6}$	$81_{\pm 9}$
Citeseer	$66.0_{\pm 0.7}$	$.59_{\pm .02}$	$.41_{\pm .02}$	$4.1_{\pm .5}$	$83_{\pm 4}$
Pubmed	$78.9_{\pm 0.7}$	$.41_{\pm .11}$	$.59_{\pm .11}$	$3.2_{\pm .4}$	$74_{\pm 23}$
Amazon	$94.8_{\pm 0.3}$	$.14_{\pm .02}$	$.86_{\pm .02}$	$3.8_{\pm .2}$	$92_{\pm 4}$
WikiCS	$77.9_{\pm 0.4}$	$.34_{\pm .04}$	$.66_{\pm .04}$	$3.8_{\pm .4}$	$70_{\pm 3}$

TABLE II: First-order envelope ratio (actual / first-order-predicted equilibrium shift) with increasing ε (10 seeds, S_c-optimal constrained perturbation). Tight at ε=0.01 (≈ 1.00); under-predicts by up to 39% at ε=0.20.

Dataset	ε=.01	ε=.05	ε=.10	ε=.20
Cora	$1.01_{\pm .00}$	$1.07_{\pm .01}$	$1.15_{\pm .03}$	$1.36_{\pm .08}$
Citeseer	$1.01_{\pm .00}$	$1.07_{\pm .01}$	$1.16_{\pm .03}$	$1.39_{\pm .07}$
Pubmed	$1.01_{\pm .00}$	$1.07_{\pm .01}$	$1.15_{\pm .02}$	$1.38_{\pm .06}$
WikiCS	$1.00_{\pm .01}$	$1.01_{\pm .01}$	$1.02_{\pm .01}$	$1.05_{\pm .01}$
Amazon	$1.00_{\pm .00}$	$1.01_{\pm .00}$	$1.03_{\pm .01}$	$1.07_{\pm .01}$

TABLE III: Equilibrium damage at ε=0.10 across attack methods (IGNN, 50-node subgraph, 10 seeds). SVD: AEGIS. Cls-PGD: classification-loss PGD. Shift-PGD: IFT-gradient PGD (solver validation, not an independent baseline).

Dataset	SVD	Cls-PGD	Shift-PGD	Random
Cora	$3.70_{\pm.79}$	$2.51_{\pm.48}$	$3.05_{\pm.70}$	$0.97_{\pm.25}$
Citeseer	$4.63_{\pm.71}$	$2.97_{\pm.42}$	$3.35_{\pm.57}$	$1.01_{\pm.15}$
WikiCS	$2.10_{\pm.22}$	$0.67_{\pm.11}$	$1.93_{\pm.20}$	$0.53_{\pm.09}$

B. Four-Quadrant Attack Comparison

Radius vs. smoothing. Randomized smoothing [7] gives constant-per-node base radii (0.123 vs. AEGIS's 0.046 at σ=0.05) and localized smoothing [9] needs 10^3–10^4 MC/node, both lacking edge structure; AEGIS radii are structurally informative (r_v≈0.09 dense, 0.01 at boundaries), a complementary point on the certificate-vs-diagnostic frontier.

Table III evaluates AEGIS against baselines spanning the four quadrants of (gradient-based, gradient-free) × (equilibrium-shift, classification-loss). The one-query SVD direction dominates: 1,000 random unit directions reach 0.45–0.49× of $\sigma_1(S_c)$; a 50-step equilibrium-shift PGD [41] using AEGIS's own IFT gradients recovers only 72–92% of its damage (a solver-validation upper bound, not an independent baseline); and a classification-loss PGD inflicts 15–70% less equilibrium damage while flipping predictions at comparable rates (0–1.8%). We accordingly term it the *maximally sensitive perturbation direction*: a first-order guarantee that empirically matches iterative attackers in one query.

Classification-gradient edges differ. Per-edge classification gradients anti-correlate with discrete ground truth on Cora ($\tau = -0.20_{\pm 0.05}$ vs. AEGIS's $+0.32_{\pm 0.04}$) and stay near zero elsewhere; AEGIS achieves 2–6× higher P@10, confirming that equilibrium and classification sensitivities are distinct vulnerability surfaces.

Discrete edge removal. See fig. 3 for four sequential orderings.

Per-edge finite differences reproduce the S_c column-norm ranking (τ=0.999); iterative S_c recomputation closes ≤ 3 pp of the static-vs-greedy gap at $k\times$ compute, so the static ranking is the headline.

Breach rates. Every breached node satisfies $\varepsilon > r_v$ across all datasets and budgets, validating r_v as a reliable first-order screening threshold (theorem 6). Figure 4 reports prediction-flip fractions across ε (bands are 95% CIs): Citeseer and WikiCS stay below 2% throughout; Cora reaches 7.6% (CI $[2.2, 13.0]$) at ε=0.20; Pubmed is the right-skewed outlier (median 7.8%, mean 10.3% at ε=0.10; 27.4%, CI $[12.3, 42.5]$, at ε=0.20).

C. Comparison with Prior Structural Baselines

Table IV reports head-to-head numbers against GR-BCD and PR-BCD [5], AGNNCert [11], and LODF [40]. AEGIS is a label-free one-query proxy; the question is how much of each gold-standard signal it retains.

GR-BCD [5] aligns with S_c on Pubmed (τ=+0.69) but decorrelates on Cora (τ=+0.16): AEGIS is a label-free one-pass diagnostic, not a budget-optimal attacker, so it is competitive rather than dominant under direct attack damage (table IV). AGNNCert [11] is complementary—a sound IBP certificate vs. first-order radii 4.9–10.2× tighter that expose the dominant direction (decision rule: table IV). Power-grid LODF/PI comparisons are in section VI.

TABLE IV: AEGIS vs. external baselines (10 seeds). GR-BCD [5] is the BCD-family scalable structural attacker (PR-BCD is the same paper's larger-budget variant, see text); AGNNCert [11] is a sound IBP certifier (semantics differ from AEGIS r_v); LODF [40] is the industry DC-PF screener.

Baseline	Setting	AEGIS	Baseline	τ
GR-BCD [5]	Pubmed k=10	0.356	0.350	**+0.69**
GR-BCD [5]	Cora k=5	0.643	1.207	+0.16
AGNNCert† [11]	Cora med. r_v	0.187	1.414	+0.08
LODF [40]	case118 P@10	0.81	$\leq$ 0.20	+0.14

†AGNNCert is a *sound* IBP certificate; AEGIS r_v is a first-order sensitivity threshold (theorem 6). Per-dataset mean tightness ratio $r_{\text{cert}}/r_v \in [4.9, 10.2]$ (per-cell across 30 seeds $[1.5, 22.5]\times$); per-seed Kendall $\tau \in [-0.08, 0.23]$. **Decision rule:** if AGNNCert certifies node v safe at budget ρ, trust it; if $r_v < \rho$ and AGNNCert does *not* certify, v is first-order unsafe and warrants attention. The methods are complementary in this prescribed sense, not interchangeable.

D. Phase Transition and Scalability

Phase transition. Figure 6 sweeps $\kappa_{\max} \in [0.30, 0.99]$ on Cora (10 seeds): even when the spectral-normalisation cap is pushed to 0.99 (driving the theoretical $\varepsilon_{\text{crit}}$ down 230×), the trained ReLU pattern keeps $\rho(J_z) \leq 0.42$, so the resolvent grows only $\mathbf{1.17{\to}1.80}$, *validating* $\varepsilon_{\text{crit}}$ as a closed-form sufficient safety boundary that holds with $\mathbf{2{-}4}\times$ empirical margin across the sweep. Training converges throughout; the matrix-free pipeline reaches its 50-step fixed-point budget at $\kappa_{\max} \geq 0.85$.

Scalability and matrix-free self-consistency. Figure 5 compares dense and matrix-free pipelines. The matrix-free $S_c v$ implements the dense S_c up to Neumann truncation; σ_1 agrees within 0.03% at N=200 (per-edge τ=0.999), and the analytical truncation residual $\kappa^{200} \in [10^{-105}, 10^{-48}]$ across the suite (consistent with the trained-IGNN range $\kappa \in [0.14, 0.59]$ of table I). For larger N the rSVD error [33] is bounded by the spectral gap; Pubmed (N=19,717) exceeds 24 GB.

E. Ablations and Defense

Subgraph size. Varying $N \in \{30, 50, 100, 200\}$ on Cora (10 seeds): tightness is stable at 1.01–1.02 for $N \leq 100$ and degrades to 1.031 at N=200; we use N=50 as default (66× faster). Subgraph-vs-full-graph agreement is high on dense grids (case14 τ=0.80–0.86).

Full-graph scale. A 50-node BFS covers only ∼1.8% of a citation graph's edges (Cora τ=0.16), so at this scale we run AEGIS on the full graph, where its edge advantage over degree-proportional *amplifies* to $\mathbf{9.82}\times$ (Citeseer) and $\mathbf{3.25}\times$ (Cora) at k=10, vs. ≈1.1× on subgraphs.

Hyperparameters. Tightness is stable across $d \in \{16, 32, 64, 128\}$; the spectral-norm ceiling $c \in \{0.5, 0.9\}$ traces the accuracy–robustness frontier (r_v=0.090@72.1% vs. 0.051@80.6%).

Defense-informed edge protection: gains survive adaptive attack. Masking the top-k edges by v_{ij} (Cora IGNN, N=50,

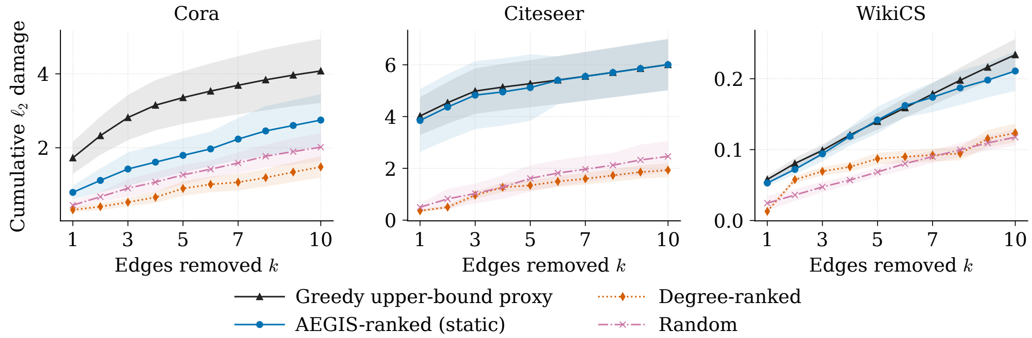


Fig. 3: Cumulative damage from sequential discrete edge removal vs. k (IGNN, 50-node subgraph, 10 seeds, ± 1 sd). AEGIS *matches* the ground-truth-aware Greedy attacker on Citeseer/WikiCS and recovers **54–67**% of Greedy's damage on Cora at k=5–10 *without any access to discrete ground truth*; degree-ranked stays below random on Cora at every k.

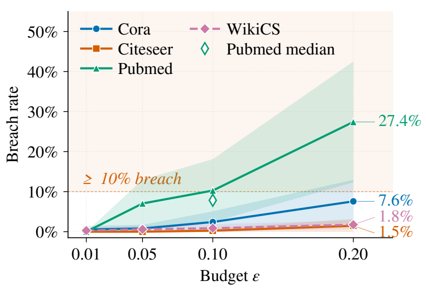


Fig. 4: Breach rate (% of nodes flipped under S_c-optimal perturbation) vs. ε (10 seeds; mean, 95% CI bands).

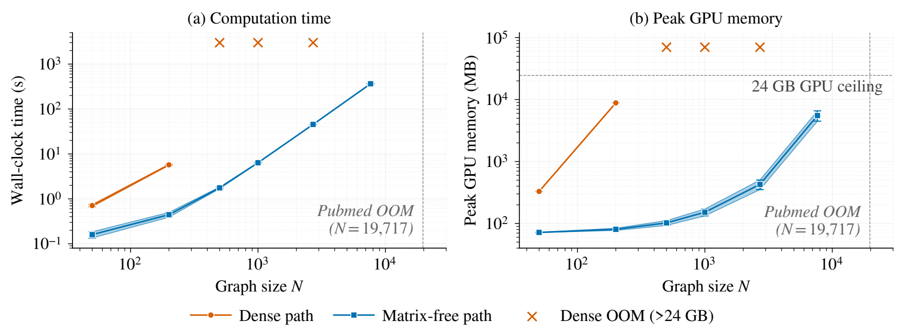


Fig. 5: Dense vs. matrix-free AEGIS pipeline (RTX 4090). Dense scales as $O((Nd)^3)$, exceeding 24 GB beyond N=200; matrix-free is roughly $O(N \log N)$ in time, sub-linear in memory, reaching Amazon Photo (N=7,650) at 365 s/5.5 GB.

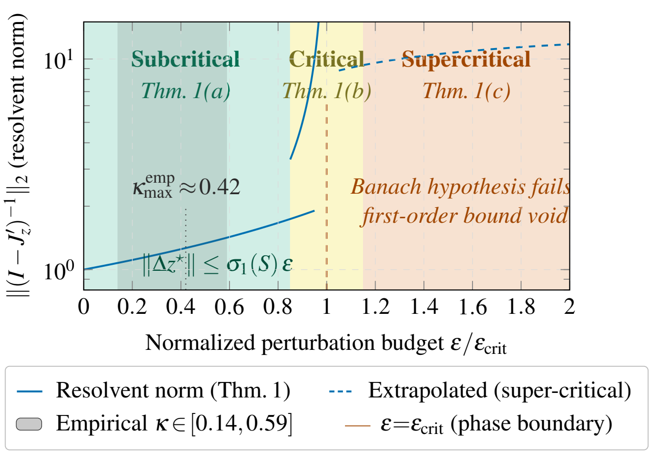


Fig. 6: Three-regime characterisation of theorem 1. The empirical κ band (grey) saturates well below the spectral-normalisation ceiling, so the empirical resolvent growth is far milder than the worst-case $1/(1-\kappa)$ divergence at $\varepsilon \to \varepsilon_{\text{crit}}$.

10 seeds) cuts $\sigma_1(S_c)$-damage by **42±8**% at k=5 and **61±7**% at k=10, vs. 11±6%/18±9% for random masking (paired Wilcoxon p<0.002). An *adaptive* attacker that recomputes S_c on the masked graph matches the non-adaptive attacker within 0.1 pp (10 seeds, ε=0.10): the 43% spectral gap of S_c (fig. 2) keeps the dominant direction robust, so the defense *does not erode under recomputation*—the failure mode sober-look critiques [16] flag.

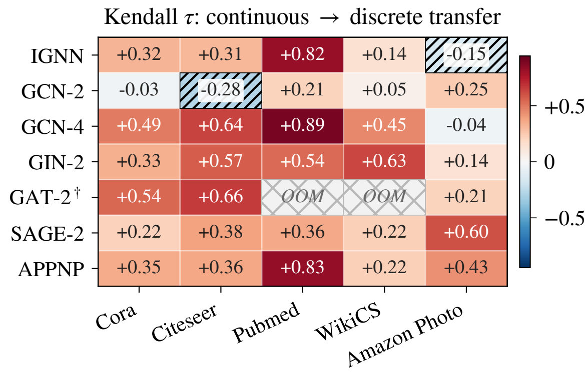
TABLE V: S_c framework on 7 GNN architectures (Cora, 10 seeds). Tight.: actual/predicted shift ratio. AtkAdv: AEGIS/random damage. τ: Kendall vs. brute-force fixed-normalization single-edge removal (theorem 7). Acc: full-graph test accuracy.

Model	Tight.	AtkAdv	τ	Acc
IGNN (eq.)	$1.01_{\pm.00}$	$7.6_{\pm0.5}\times$	$+.32_{\pm.04}$	$77.5_{\pm1.7}$%
GCN-2	$1.00_{\pm.00}$	$3.6_{\pm0.6}\times$	$-.04_{\pm.03}$	$78.9_{\pm0.9}$%
GCN-4	$1.02_{\pm.00}$	$3.4_{\pm0.4}\times$	$+.49_{\pm.02}$	$78.3_{\pm1.8}$%
GIN-2	$1.01_{\pm.00}$	$2.6_{\pm0.1}\times$	$+.33_{\pm.03}$	$76.6_{\pm1.9}$%
GAT†	$1.01_{\pm.01}$	$2.1_{\pm0.1}\times$	$+\mathbf{.54}_{\pm.06}$	$77.8_{\pm1.2}$%
SAGE-2	$1.00_{\pm.00}$	$1.9_{\pm0.2}\times$	$+.22_{\pm.10}$	$77.5_{\pm1.5}$%
APPNP	$1.02_{\pm.00}$	$3.4_{\pm0.3}\times$	$+.35_{\pm.01}$	$\mathbf{82.2}_{\pm0.6}$%

F. Extension to Explicit GNNs

For K-layer explicit GNNs (GCN, SAGE, GIN, APPNP, GAT† [42], [43], [44], [45], [46]) we form S_K=$\partial\text{vec}(Z_K)/\partial\text{vec}(A)$ via finite differences and construct S_c from S_K identically (theorem 8). Table V reports the per-architecture characterisation on Cora: IGNN carries theorem 1; the explicit models receive the computational tool without the regime characterisation.

GAT† scope. Standard GAT uses A as a binary mask, so $\partial Z/\partial A_{ij}$=0 on existing edges and S_c is undefined; our GAT†



† *GAT-2: 2-layer attention; OOM on Pubmed and WikiCS.*

Fig. 7: Continuous-to-discrete transfer τ across 7 architectures and 5 datasets (10 seeds, 330 runs). Cell values are mean Kendall τ; buckets: strong (τ>0.5), moderate (0.3–0.5), weak (0–0.3), anti-correlated (τ<0, diagonal hatch); cross-hatch: GPU OOM. Deeper models (GCN-4, APPNP) and GAT† dominate.

modulates attention by the edge weight ($h'_i = \sum_j \hat{A}_{ij}\alpha_{ij}Wh_j$), restoring differentiability and yielding tightness 0.99, AtkAdv $4.4\times$, $\tau = +0.36$. Binary-mask architectures (hard attention, max/min aggregation, GATv2 [47]-style dynamic-attention masking) fall outside the framework.

Robust backbones. Under the $\sigma_1(W) \leq 0.9$ cap, RobustGCN-lite [13] ($\tau = +0.37/+0.54$, Cora/Citeseer) and GNNGuard-lite [14] ($+0.10/+0.53$) match or exceed IGNN; the cap is a precondition (without it $\kappa > 1$ voids theorem 1).

Cross-dataset transfer. Figure 7 reports τ over 5 datasets and 7 architectures (330 runs); $29/33$ cells positive (one-sided sign test $p < 10^{-5}$). Cold cells (GCN-2 on sparse citation graphs, IGNN on dense product graphs at 50 nodes) recover decisively at full-graph scale: on Amazon Photo (N=7,650, stratified top-v_{ij} ground-truth sampling) the edge-weighted ranking $A_{ij}v_{ij}$ predicted by theorem 7 reaches $\tau = \mathbf{+0.996 \pm 0.002}$ (10 seeds) against brute-force fixed-normalization N-1 damage, a near-perfect rank match across seeds. On heterophilic WebKB [48] GCN-4 attains $\tau = +0.21$–$+0.44$.

Theorem 7's sufficient condition holds for **47–62**% of edge pairs across architectures; GCN-2's 2-hop aggregates produce near-uniform first-order shifts and fall outside the sufficient regime, which the framework correctly flags via a low dataset-level τ (fig. 7).

VI. Case Study: Power Flow Contingency

Beyond classification, AEGIS extends to AC power-flow contingency analysis: applied to a learned ContractiveGCN-PF surrogate, the S_c ranking recovers brute-force N-1 critical-line severity on IEEE case14–118 (τ=+0.37 to +0.62, **P@10=0.66–0.81**), competitive with industry LODF screening. The ranking is label-free and one-query: v_{ij} comes purely from the surrogate's learned IFT-resolvent $(I - J_z)^{-1}$, and its agreement with PTDF/LODF [40] is consistent with the topology-driven flow concentration those DC measures encode. Recent GNN-PF work targets fidelity [49], [50], [51], [3] and contingency screening [52]; AEGIS adds a label-free vulnerability-attribution layer over any such surrogate, without contingency labels or retraining.

TABLE VI: AEGIS on IEEE power-flow benchmarks (10 seeds). Model quality: per-unit RMSE for $|V|$ and angle θ (std < 0.003). τ: Kendall correlation against brute-force N-1; P@10: top-10 precision.

Case	N	$\|E\|$	Model (p.u.) $\|V\|$	Model (p.u.) θ	AEGIS τ	AEGIS P@10
case14	14	20	.007	.020	$+.42_{\pm.19}$	$.74_{\pm.12}$
case30	30	41	.014	.024	$+.37_{\pm.17}$	$.68_{\pm.06}$
case57	57	78	.033	.059	$+.67_{\pm.09}$	$.66_{\pm.13}$
case118	118	179	.014	.076	$+.62_{\pm.11}$	$.81_{\pm.10}$

Setup. ContractiveGCN-PF is an IGNN-class operator $F(Z, A) = \phi(\hat{A}ZW^\top + X_{\text{proj}})$ with spectral-norm-constrained W, trained on five IEEE test cases [38] with a 2-head readout $(\Delta|V|, \Delta\theta)$. Training data are 2,000 load samples per case (70–130% of nominal, *uniform load scaling only*, a narrow envelope that does not cover seasonal peaks, dispatch shifts, or renewable ramps) from PandaPower's [39] AC Newton-Raphson solver; inputs are binary adjacency and 5 bus features (slack/PV indicators, P, Q, $|V|$). Ground truth: ℓ_2 voltage-angle deviation per AC contingency. We use full-graph dense analysis on case14–118. LODF's native metric is DC line-flow change; we benchmark its rank-order against AC voltage-angle truth, where the comparison stays informative within LODF's scope.

table VI reports the envelope ratio and Kendall τ between S_c rankings and brute-force N-1; fig. 8 visualises case14. The envelope is tight in the PF domain (≈ 1.00), and on case14–118 the ranking correlation τ=+0.37 to +0.62 (P@10=0.66–0.81) identifies 7–8 of the top-10 critical lines. The matrix-free S_c pipeline scales to N=7,650 (fig. 5); richer PF surrogates trained on broader operating envelopes are the path to operational-grade screening.

Baselines. Against the AC voltage-angle truth, LODF (industry DC screening) attains τ=0.44–0.58 on the credible cases; AEGIS matches or leads it (0.62–0.67 on case57/118, Wilcoxon p<0.01; overlapping on case14/30), and on LODF's fairer thermal-overload target reaches P@10=0.60 (case57)—competitive, not dominant. *Binary adjacency beats admittance-weighting* (P@10=0.81 vs. 0.27, case118): a single-line trip removes the line regardless of impedance.[1]

VII. Related Work

We position AEGIS against three threads; each supplies at most one of the three diagnostics AEGIS produces from one S_c object via a single matrix-free pipeline.

Structural attacks. Nettack [1], Mettack [4], GR-BCD/PR-BCD [5] and variants [6] optimise surrogate gradients to flip

[1]Other baselines (case57/118): Ejebe–Wollenberg PI [53] τ=+0.335/+0.101 (P@10=0.50/0.30); standalone PTDF without outage correction is anti-correlated/near-zero (τ=−0.14/+0.06), confirming outage redistribution is the operative effect. Runtime: LODF <0.13 s, N-1 0.1–2 s, AEGIS 2–23 s. One N-1-targeted pass recovers 40–64% of N-2 critical sets (top-5 overlap; pairwise stability +0.40–+0.78).

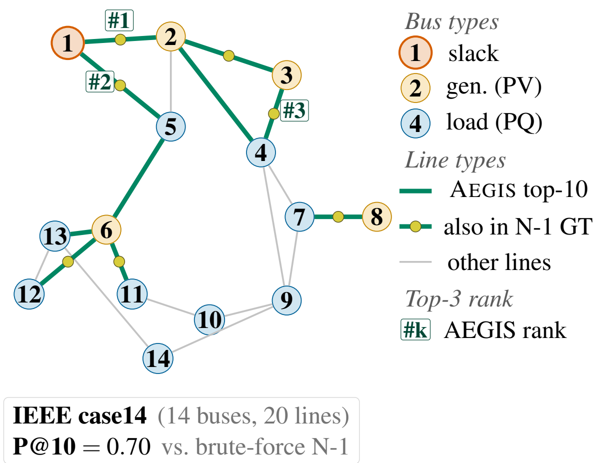


Fig. 8: AEGIS vulnerability ranking on the IEEE 14-bus benchmark (10-seed mean; P@10 $= 0.70$ for this run, consistent with the 0.74 ± 0.12 in table VI). Bus markers encode physical bus type (slack / generator / load); green edges are the top-10 by v_{ij} with rank tags (#1–#3); yellow midpoint dots mark edges that also appear in the brute-force N-1 top-10 (7 of 10). Top-ranked edges concentrate on the high-flow inter-area tie lines (1–2, 1–5, 7–8), recovering the single-contingency severity pattern from one matrix-free SVD pass.

predictions but provide no analytic globally maximally sensitive direction and no per-node radius. Sober-look studies [16], [17] flag evaluation pitfalls; we follow their adaptive-attack discipline in section V-E (paired Wilcoxon against random masking; an adaptive-attacker column in defense ablation).

Certified robustness and defense. Convex relaxations [12], smoothing [7], [8], [9], [54], collective certificates [10], and IBP [11] certify per-node robustness but do not identify vulnerability-driving edges. Empirical defenses [13], [15], [14] harden models without surfacing per-edge sensitivity. r_v is a first-order threshold, not a sound certificate (theorem 6); the methods are complementary under the prescribed decision rule of table IV.

Implicit networks and influence functions. DEQs [20], IGNN [22], monotone equilibria [55], and well-posedness analyses [28] address *input* sensitivity $\partial z^*/\partial x$; AEGIS targets *structural* sensitivity $\partial Z/\partial A$ via the resolvent $(I - J_z)^{-1} J_A$ and the unrolled Jacobian S_K. The construction descends from the Koh–Liang influence-function lineage [18] and from implicit-differentiation hyperparameter optimisation [27], [19], but those lines apply IFT to feature- or weight-perturbations of a loss; AEGIS applies it to a *structural-parameter* perturbation of the equilibrium state, projects onto the edge subspace via P_c, and obtains the SVD-optimal direction in a single matrix-free pass rather than via iterative reverse-mode. GNN explainers [56] target prediction fidelity (which subgraph rationalises an output), not adversarial vulnerability; classical matrix perturbation theory [31], [26] supplies the pseudospectral toolkit.

VIII. Conclusion

AEGIS produces three adversarial diagnostics (per-edge rankings, the SVD-optimal attack direction, per-node first-order sensitivity radii) from one analytical object, the constrained sensitivity matrix S_c, computed matrix-free for any GNN with continuous edge-weight-modulated message passing, with a regime characterisation (theorem 1) that holds as a sufficient safety boundary for contractive implicit models.

Limitations. (i) Radii r_v are first-order thresholds, not probabilistic certificates (theorem 6); standard GAT and binary-mask architectures fall outside the framework (section V-F). (ii) Full-graph matrix-free analysis is the recommended graph-level path: AtkAdv amplifies ($3.3\times/9.8\times$ on Cora/Citeseer) and the edge-weighted ranking $A_{ij}v_{ij}$ achieves $\tau = +\mathbf{0.996}$ on Amazon Photo (N=7,650; section V-F). (iii) Opting into the formal $\varepsilon_{\text{crit}}$ track accepts a $\sim 6\%$ accuracy cost, on par with certified-robustness trade-offs. (iv) Operator-grade power-flow deployment, extending case14–118's τ=+0.37–+0.62 to admittance-weighted edges and broader envelopes, is a natural follow-up. (v) Insertion attacks (Nettack-class) are deliberately scoped out; the S_c construction extends once the edge basis is enlarged to $\bar{E} \supseteq E$ (section II).

Disclosure protocol. (i) A 90-day coordinated-notification window before public release, addressed to the maintainers of OGB graph-robustness benchmarks and to the GR-BCD / Mettack / AGNNCert authors. (ii) Attack-direction code (Algorithm 1 steps 8–9) gated behind institutional-affiliation review mediated by an institutional research-ethics office (named in the camera-ready). (iii) The diagnostic-only path (r_v and v_{ij} scores; no SVD reconstruction) released unconditionally, since it cannot directly synthesise a perturbation. Power-grid artefacts use only public IEEE benchmarks [38].

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